Yuhang Sun

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EDUCATION

University of Washington, Global Innovation Exchange (GIX) Seattle, WA
M.S. in Technology Innovation, Robotics Track. GPA 3.8/4.00 *Sep 2025 – Present*

Wuhan University, School of Urban Design Wuhan, China
B.A. in Product Design. GPA 3.81/4.00 *Sep 2021 – Jun 2025*

University of California, Berkeley Extension Berkeley, CA
Berkeley International Study Program. GPA 3.81/4.00 *Jan 2024 – May 2024*

RESEARCH INTERESTS

physical interaction and tactile sensing
safety and reliability
immersive VR and behavioral measurement
robot learning and embodied AI
world models and spatial generalization
digital fabrication and interactive devices

PUBLICATIONS AND MANUSCRIPTS

[1] Yuhang Sun. **Physical Experience Builds Continual Spatial World Models.** NeurIPS 2026 Workshop on Continual World Models | Accepted

[2] Yuhang Sun. **Action Representation for Spatial Generalization in Robot World Models**. Robotics and Autonomous Systems, 2026 | Submitted; with editor.

RESEARCH EXPERIENCE

Benchmarking World Model Reliability Under Environmental Perturbations University of Washington
Student Researcher | Advisor: Luyao Niu *Sep 2026 – Present*

- Design a multi-environment benchmark for evaluating the robustness and reliability of learned world models under controlled environmental perturbations, focusing on prediction consistency, model generalization, and false-safe assessments.
- Develop checkpoint-based evaluation protocols comparing predicted and ground-truth trajectories under matched initial states and action sequences, incorporating systematic perturbations to characterize world-model prediction failures.

Auxetic Structures for Robotic Tactile Sensing University of Washington
Student Researcher | UbiComp Lab | Mentor: Alex Ching *Sep 2026 – Present*

- Investigate auxetic structures with negative Poisson's ratios for flexible tactile sensing on robotic surfaces, focusing on mechanical deformation, conformability, and integration with contact sensing components.
- Design and experimentally evaluate mechanical structures for robotic tactile sensing, exploring geometric configurations and their feasibility for integration into deformable sensing surfaces.

Capacitive Insole Sensing for Foot Pressure Measurement University of Washington
Student Researcher | UbiComp Lab | Mentor: Alex Ching *Jul 2026 – Present*

- Explore capacitive sensing approaches for wearable foot pressure measurement, investigating how sensor configurations and mechanical deformation influence capacitance-based sensing responses.
- Conduct preliminary capacitive sensor experiments and develop insole models to examine sensor placement, structural integration, and the feasibility of wearable pressure sensing systems.

Physical Experience Builds Continual Spatial World Models University of Washington
Independent Research *May 2026 – Present*

- Investigate how embodied agents acquire and continuously update spatial world models through physical interactions, focusing on transferable physical relationships that enable predictions across previously unvisited locations.
- Develop a physics-grounded spatial modeling framework in maze environments, using interaction-driven parameter estimation and continual model updates to predict environmental dynamics across different spatial configurations; resulting paper accepted at a NeurIPS 2026 workshop.

SafeWorld: Temporal Safety Verification for Learned World Models University of Washington
Student Researcher | Advisor: Luyao Niu *Apr 2026 – Present*

- Investigate the reliability of learned world models in safety-critical decision making by combining action-conditioned latent predictions with formal temporal specifications based on Linear Temporal Logic (LTL) and Signal Temporal Logic (STL).
- Integrate trained SafeDreamer checkpoints into a temporal verification framework, implementing safety evaluation procedures to assess predicted trajectories and examine discrepancies between model-based safety assessments and actual environment behavior.

Attention and Mind Wandering in Virtual Reality Classrooms University of Washington
Student Researcher | NEWT Lab | Mentor: Jiexin Ding *Mar 2026 – Present*

- Investigate attention, mind wandering, and learning performance in immersive VR classrooms, leveraging eyetracking and behavioral data collected through Meta Quest Pro to characterize students' cognitive states during learning activities.

- Develop interactive VR classroom environments in Unity and contribute to experimental design, multimodal data collection, and behavioral analysis, conducting pilot studies to refine experimental protocols and assess feasibility.

WORK EXPERIENCE

Global Innovation Exchange (GIX) University of Washington

TECHIN 511: Fabrication & Physical Prototyping Teaching Assistant *Sep 2026 – Present*

- Support teaching and grading for a graduate course in fabrication and physical prototyping for interactive systems.

SERVICE

Conference Reviewer: NeurIPS 2026 Continual World Models Workshop

SKILLS

Programming & Tools: Python, MATLAB, Git, C, C#, Unity
Robotics: LeRobot, ROS2, MoveIt2, RViz, TF2
Modeling & Prototyping: CAD, Rhino, Blender, SolidWorks, 3D printing
Languages: Mandarin (Native), English